

Unit 8, Lesson 4: Graphing Quadratic Equations From Key Features



Benchmarks

MA.912.AR.3.6 Given an expression or equation representing a quadratic function, determine the vertex and zeros and interpret them in terms of a real-world context.

MA.912.AR.3.7 Given a table, equation, or written description of a quadratic function, graph that function and determine and interpret its key features.



Learning Targets

- ☐ I can determine key features of a quadratic function given equations in different forms.
- ☐ I can graph a quadratic function using key features determined from equations and tables of values.



8.4.1 Warm-Up: How's the Weather?

The graph shows solar radiation, S , in watts per square meter, as a function of time, t , in hours since midnight. Think about this quantity as the maximum amount of power that a solar panel can absorb, which is determined by how intense the sunshine is at any given time.

1. Which description corresponds to the graph of the weather during the day?



Hours Since Midnight

Descriptions

- | | |
|----|--|
| A. | It was a beautifully sunny day from sunrise to sunset – not a cloud in the sky. |
| B. | The day started off foggy but eventually the fog lifted, and it was sunny the rest of the day. |
| C. | It was a pretty gloomy day. The morning fog never really lifted. |

2. Compare with your partner and discuss how you determined the description that matched the graph.



8.4.2 Guided Instruction: Graphing Quadratic Equations From Key Features and a Table of Values

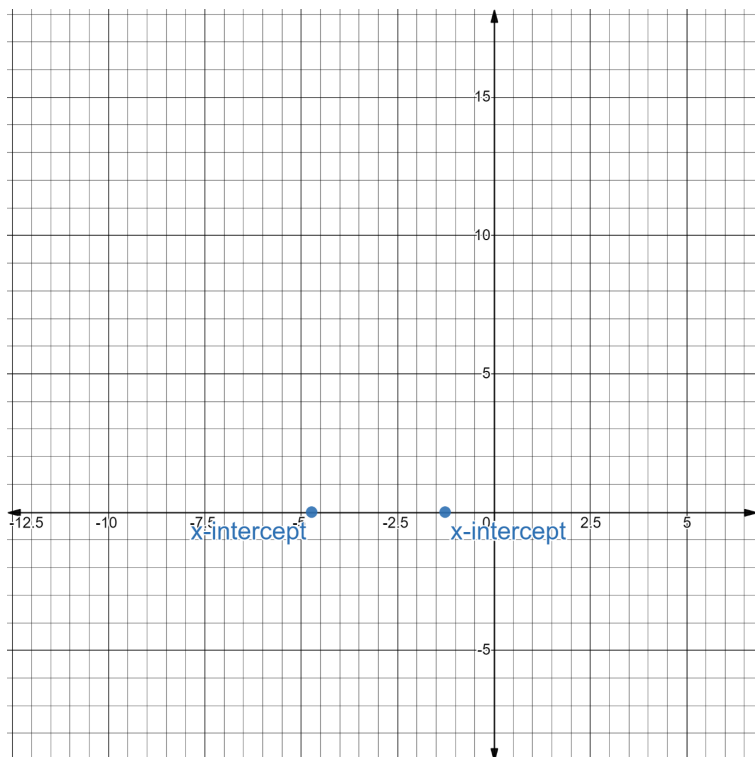
The three forms of quadratics – vertex form, factored form, and standard form – each highlight key features of a quadratic relationship that can assist in graphing.

1. Consider the quadratic equation $y = 2(x + 3)^2 - 6$.
 - a. Write the key features that can be determined from the equation $y = 2(x + 3)^2 - 6$ in the table.

Key Features			
x -intercept		increasing	
x -intercept		decreasing	
y -intercept		positive	
vertex		negative	
domain		opens up/down	
range		axis of symmetry	
left end behavior	$x \rightarrow -\infty,$ $y \rightarrow \underline{\hspace{1cm}}$	right end behavior	$x \rightarrow \infty,$ $y \rightarrow \underline{\hspace{1cm}}$

- b. Rewrite the equation $y = 2(x + 3)^2 - 6$ in standard form.
 - c. Add any additional key features that can be determined from the equation in standard form to the table above.

The x -intercepts of $y = 2(x + 3)^2 - 6$ are plotted on the graph.



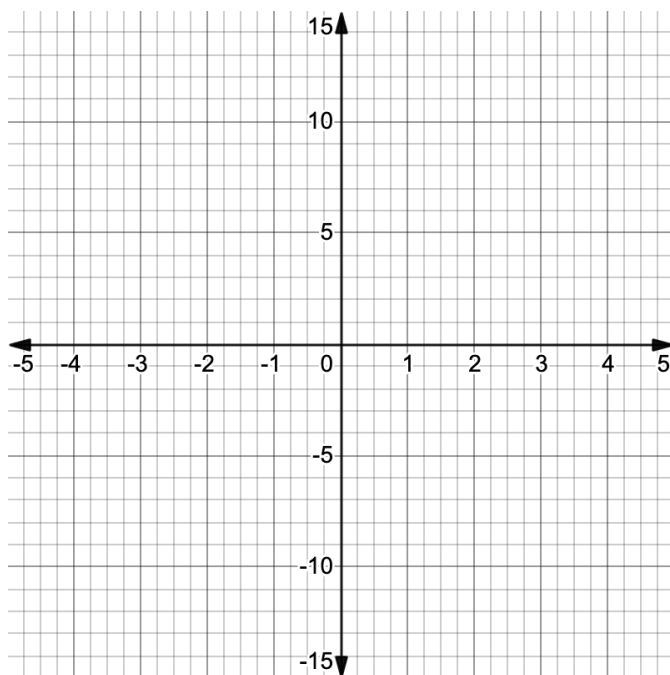
- d. Plot and label the vertex and the y -intercept. Then, draw the parabola through the key features plotted.
- e. Discuss with your partner if there are additional key features that can be determined from the graph. If so, add them to the key features table.

2. Consider the equation $y = -4(x - 1)(x - 3)$.

- a. Write the key features that can be determined from the equation $y = -4(x - 1)(x - 3)$ in the table.

Key Features			
x -intercept		increasing	
x -intercept		decreasing	
y -intercept		positive	
vertex		negative	
domain		opens up/down	
range		axis of symmetry	
left end behavior	$x \rightarrow -\infty,$ $y \rightarrow \underline{\hspace{1cm}}$	right end behavior	$x \rightarrow \infty,$ $y \rightarrow \underline{\hspace{1cm}}$

- b. Plot and label the x -intercepts of $y = -4(x - 1)(x - 3)$.



- c. Complete the table of values using the equation.

x	y
0	
2	
4	

- d. Use the x -intercepts and the ordered pairs in the table to graph the parabola.
- e. Discuss with your partner if there are additional key features that can be determined from the table and graph. If so, add them to the key features table.

3. Consider the equation $y = 5x^2 - 20x + 15$.

- a. Write the key features that can be determined from the equation $y = 5x^2 - 20x + 15$ in the table.

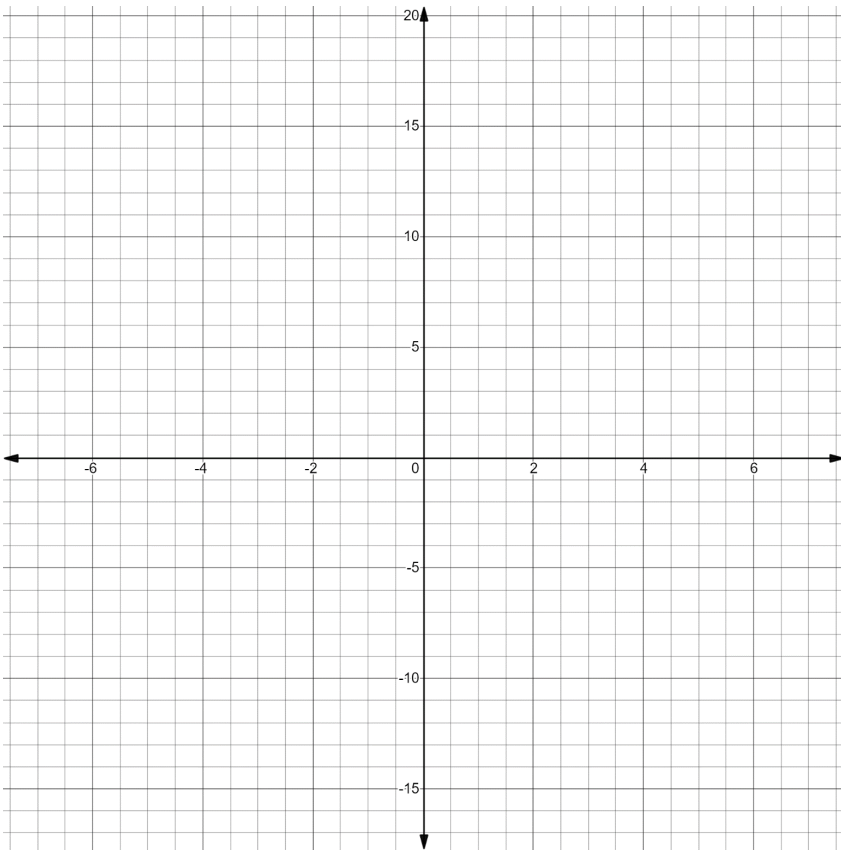
Key Features			
x -intercept		increasing	
x -intercept		decreasing	
y -intercept		positive	
vertex		negative	
domain		opens up/down	
range		axis of symmetry	
left end behavior	$x \rightarrow -\infty,$ $y \rightarrow \underline{\hspace{1cm}}$	right end behavior	$x \rightarrow \infty,$ $y \rightarrow \underline{\hspace{1cm}}$

- b. Rewrite the equation $y = 5x^2 - 20x + 15$ in factored form.

c. What additional key features are revealed from the equation in factored form?

d. Add any additional key features revealed from factored form to the table.

e. Graph $y = 5x^2 - 20x + 15$.



f. Label the intercepts and the vertex.

Equations and tables of values are useful tools to use when graphing quadratic relationships. Equations can be rewritten in different, equivalent forms to reveal key features. Tables of values can also be used to identify key features as needed.

4. To graph a quadratic relationship, which key features are helpful to have?

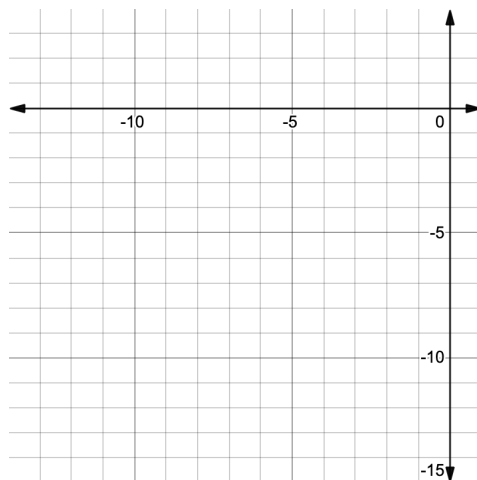


8.4.3 Guided Practice: Graphing Quadratic Equations From Key Features and a Table of Values

1. Consider the equation $y = -\frac{1}{3}(x + 6)^2$.

- a. Create a table of values.

x	y



- b. Use key features and a table of values to graph $y = -\frac{1}{3}(x + 6)^2$.

- c. Work with your partner to complete the statement.

The equation $y = -\frac{1}{3}(x + 6)^2$ results in the

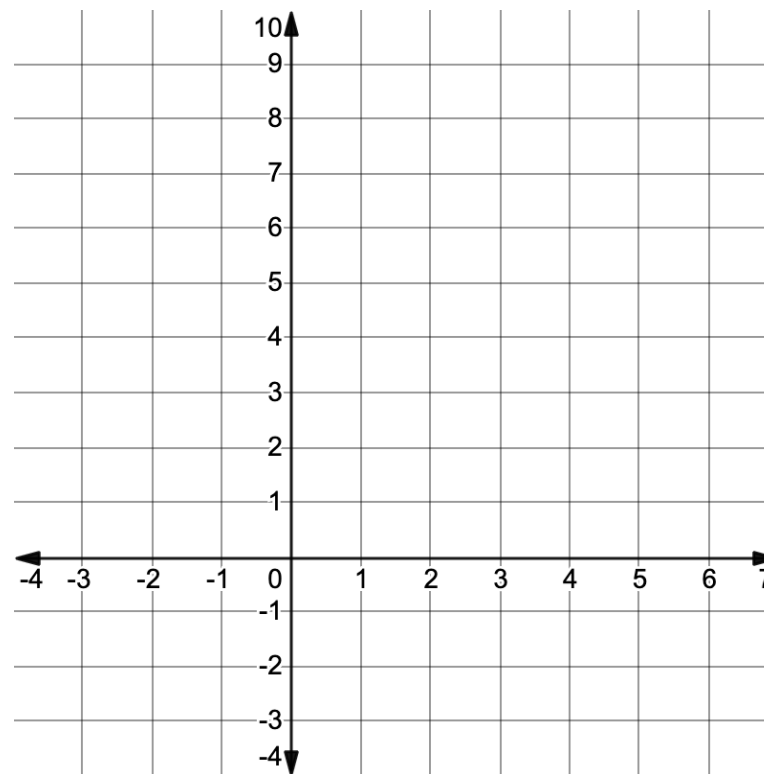
- ☐ x -intercept and the vertex being the
☐ x -intercept and the y -intercept

same because

- ☐ the a -value in the equation is a fraction.
☐ the k -value in the equation is 0.

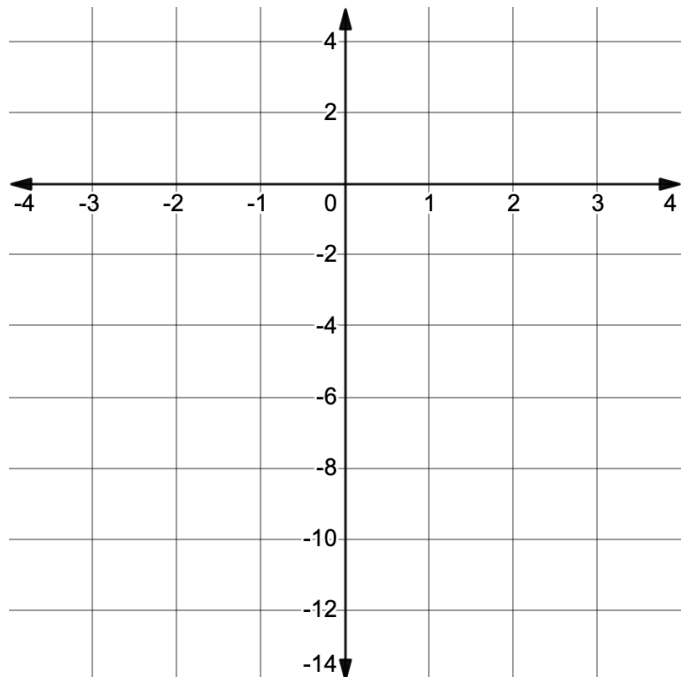
2. The vertex of the graph of $y = -(x - 5)(x + 1)$ is $(2, 9)$.

Graph the parabola using the vertex and other key features.



3. Consider the quadratic function $f(x) = 3x^2 - 12$.

a. Plot the y -intercept of the equation $f(x) = 3x^2 - 12$.



b. Rewrite $f(x) = 3x^2 - 12$ in factored form. Then, graph the x -intercepts.

c. Discuss with your partner why plotting the two x -intercepts and the y -intercept is enough information to plot a parabola that shows all of the key features for $f(x) = 3x^2 - 12$.



8.4.4 Your Turn!

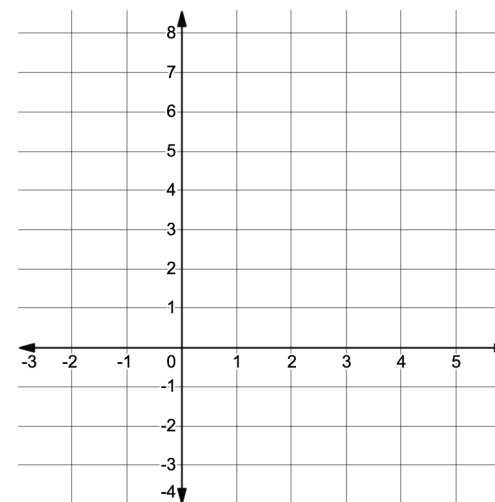
Determine the key features revealed for each quadratic equation, then complete the table of values.

Use the key features and the table of values to graph each quadratic equation.

1. $y = (x - 2)^2 - 1$

Key Features:

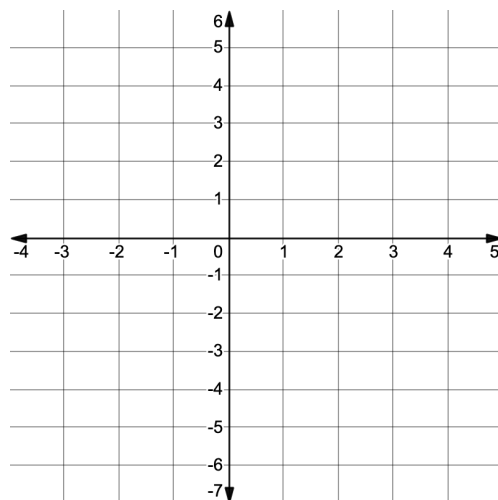
x	y
0	
1	
2	
3	
4	
5	



2. $y = -(x + 1)(x - 3)$

Key Features:

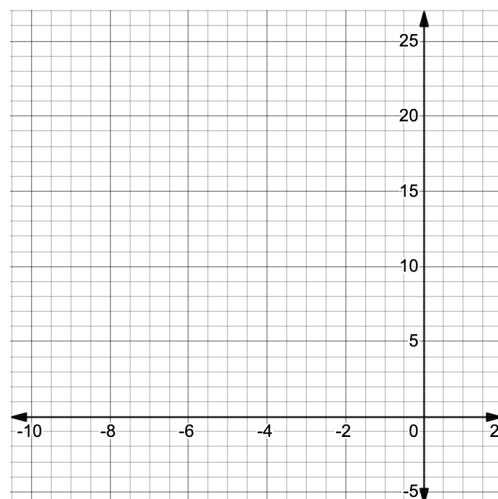
x	y



3. $y = (x + 5)^2$

Key Features:

x	y

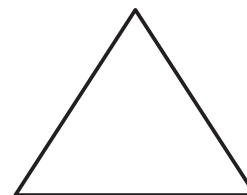


8.4.5 Wrap-Up: Reflection

1. What is something that is now “squared away” in your mind about graphing quadratics and key features?



2. What are three “points” that are important about graphing quadratics and key features?



3. What is something that is still “circling” your brain about graphing quadratics and key features?

